

Exercice 5
Advanced Computer Architecture

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1 Introduction

In this lab session we looked at the performance effect on synchronous vs asynchronous GPU-execution. We timed both of them and compare their execution time.

2 Method

In this lab we did 4 different operations on array's, the array's are generated just before the operation is executed. The operations on the arrays happen on the GPU, while the initialization of the arrays happen on the CPU. There are two main ways of interleaving these operations. First we could wait for the operation on the GPU to end before starting filling the next array on the CPU. This is called the synchronous approach, the CPU and GPU wait on each other to finish. To achieve this we need to synchronize the CPU and GPU after we call the GPU function (cudaDeviceSynchronize).

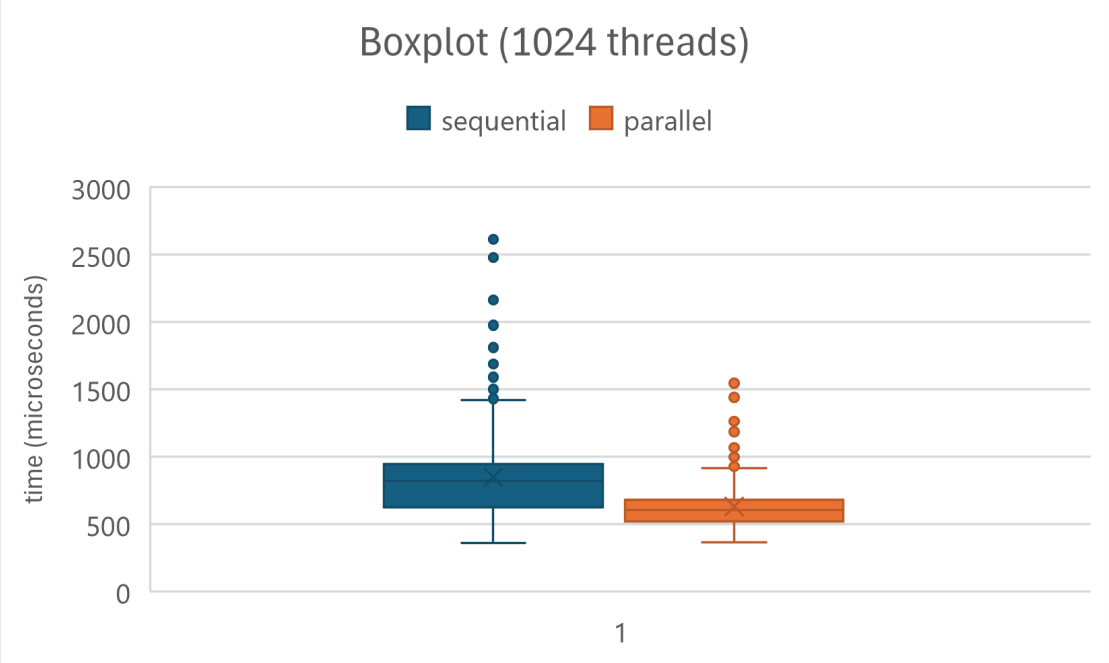
Alternatively we could start filling the next array (using the CPU) just after the GPU is evoked. This means we don't wait for the GPU to finish its operation on the first array before starting to fill the second. This can of course only be done when the second array is not dependent on the result of the GPU function (which in this case it is not). To achieve this, we need to use cudaMallocAsync.

Note that we can only use one block in this exercise because if not the blocks will wait on each other to finish, which may put noise on our measurements. Since we are using reduction on our arrays, the amount of threads per block is always the size of the array divided by two.

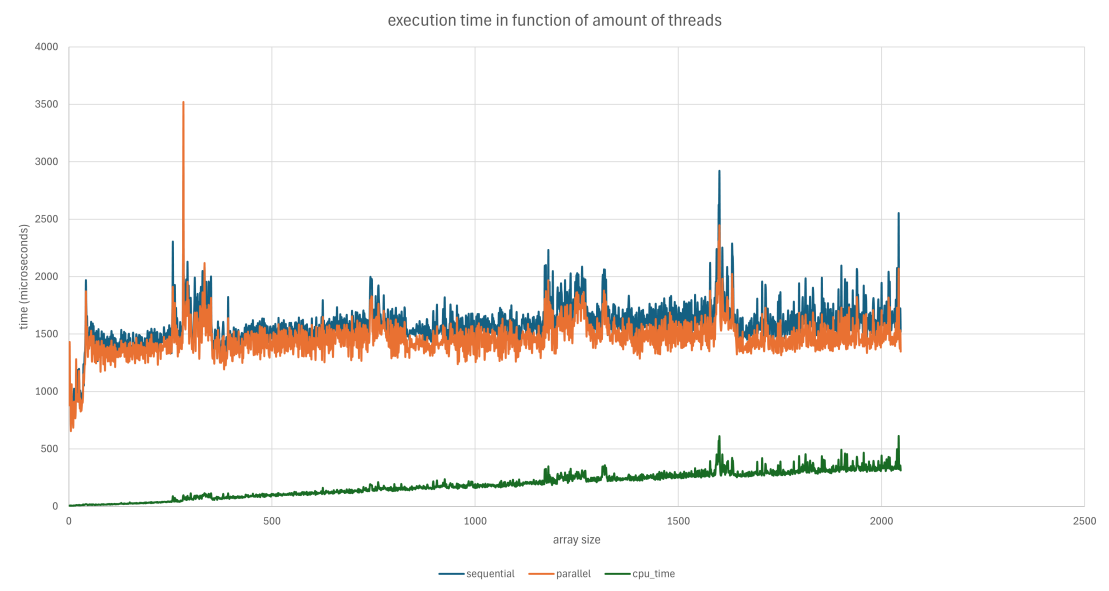
3 Observations

In figure 1 we see a boxplot of the execution time after running our synchronous and parallel implementation for 400 times, with 100 warmup rounds. We used an array of 2048 elements. From this boxplot we conclude that the synchornous method is indeed slower, and also less consistent. The asynchronous method is very consistent, from this we conclude that the execution time on the GPU is quite variable, compared to the CPU.

In figure 2 we compare the execution times for the synchronous and asynchronous methods in function of the arrays size. From this figure we conclude that the asynchronous method is faster, but the difference is does not change drastically in time. This is because the time needed for the operations on the GPU don't change much when the array gets bigger (because of parallelism). The execution times do go up when the array gets bigger because of the CPU component (the green line), the filling of the arrays take longer when the array gets bigger. The time spent on the CPU is rather small compared to the time spent on the GPU, so the influence is not that big.



Figuur 1: Execution time with size of 2048



Figuur 2: Speed comparison